Social Isolation and Subjective Cognitive Decline in 192,567 U.S. Adults Aged 50+: An Interpretable Machine Learning Screening Study Using BRFSS 2023–2024

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Abstract

Background: Social isolation and loneliness confer a 23–72% elevated dementia risk in large meta-analyses [1, 2], yet no machine learning (ML) screening tool has been built for older U.S. adults at population scale using multi-dimensional social isolation features from routinely collected survey data.

Objective: To develop and validate an interpretable ML model predicting subjective cognitive decline (SCD) from multi-dimensional social isolation and health indicators in a nationally representative U.S. cohort, and to quantify operational screening metrics.

Methods: Cross-sectional ML study using Behavioral Risk Factor Surveillance System (BRFSS) 2023–2024 data. Adults aged ≥50 with complete Cognitive Impairment and Memory Loss module responses were included (N = 192,567; SCD prevalence 17.0%, n = 32,655). We engineered 33 predictors including a novel Multidimensional Social Isolation Index (MSII). Five classifiers — logistic regression, random forest, XGBoost, LightGBM, and a stacking ensemble — were trained on the 2023 wave (n = 121,970) and evaluated on the 2024 holdout wave (n = 70,597). SHAP values quantified feature contributions. The DeLong method estimated 95% confidence intervals for AUROC.

Results: XGBoost achieved the highest holdout AUROC of 0.7678 (95% CI 0.7595–0.7761), AUPRC of 0.4339, and Brier score of 0.1193. At the optimal threshold (0.1697), sensitivity was 0.72, specificity 0.68, PPV 0.31, NPV 0.92, and the number needed to screen (NNS) was 3.2. The top SHAP predictors were depression severity, poor mental health days, and general health status; the MSII isolation composite ranked 7th. Subgroup AUROC ranged from 0.70 (age ≥80) to 0.82 (age 50–64). LightGBM performed comparably (AUROC 0.7669).

Conclusions: An XGBoost model trained on routine BRFSS social and health data detects SCD with AUROC 0.77, offering a no-biomarker population-level screening tool with high NPV (0.92). At N = 192,567, this is among the largest U.S. BRFSS ML studies focused on multi-dimensional social isolation features for SCD screening; results are consistent with the concurrent BRFSS benchmark by Nguyen et al. [3] (N = 298,944, AUROC ≈0.76–0.77). Depression co-morbidity dominates prediction;

the MSII adds incremental signal (ablation ΔAUROC = +0.006, NRI = 0.218). The model's NNS of 3.2 supports its utility as a first-tier screening instrument for primary care social prescribing pathways.

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1 Introduction

Dementia affects approximately 55 million people worldwide, with projections exceeding 150 million by 2050 [1]. Subjective cognitive decline (SCD) — self-reported worsening of memory or confusion — is an established preclinical marker that precedes clinically detectable cognitive impairment by years and carries a substantially elevated conversion risk to Alzheimer's disease and related dementias (ADRD) [4]. Identifying individuals at risk for SCD before objective impairment emerges offers a critical window for preventive intervention.

Social isolation and loneliness have emerged as powerful, modifiable risk factors for cognitive decline. A 2022 meta-analysis of 16 cohort studies (N = 62,345) by Qiao et al. found that loneliness was associated with a relative risk of 1.72 (95% CI 1.32–2.23) for Alzheimer's disease and 1.23 (95% CI 1.16–1.31) for any dementia [1]. A larger meta-analysis by Penninkilampi et al. (n = 2,370,452) reported relative risks of 1.38 for loneliness and 1.59 for poor social networks [2]. In the prospective Health and Retirement Study (N = 12,030), Sutin et al. documented approximately 40% increased dementia risk among lonely older adults over 10 years [5]. Choi et al. demonstrated that neighborhood disorder elevates dementia risk partly through subjective social isolation as a mediator (CHR 1.27, 95% CI 1.03–1.59) [6].

Despite this compelling epidemiological evidence, the translation from *association* to *operational screening tool* has not been achieved for U.S. older adults at population scale. Fasokun et al. (2026) used BRFSS data (N = 86,520) to document a strong dose-response between loneliness frequency and SCD prevalence (45.7% vs. 9.9% for always-vs.-never lonely), but reported no ML model, no AUROC, and no actionable screening threshold [7]. The most directly comparable prior work is Nguyen et al. [3], who applied gradient-boosted trees to pooled BRFSS data (N = 298,944; AUROC ≈0.76–0.77), establishing a BRFSS-specific performance ceiling of approximately AUROC 0.77. The present study contributes a distinct thematic focus: the first BRFSS ML analysis to operationalize a multi-dimensional social isolation construct (the MSII) as the primary exposure framework, quantify its incremental contribution via ablation, and report a fully actionable

screening threshold with NNS. Clinical-cohort ML studies — Ishikawa et al. (NSHAP, N = 4,586; AUROC 0.823) [8] and Alam et al. (NHATS, N = 5,984; AUROC 0.823) [9] — achieved higher AUROCs using richer clinical features but at sample sizes more than 30-fold smaller. All-of-Us-based studies [10, 11] approach comparable scale but focus on composite outcomes or racial disparity characterization rather than a dedicated social-isolation screening tool.

The Behavioral Risk Factor Surveillance System (BRFSS) is the world's largest ongoing telephone health survey, interviewing more than 400,000 U.S. adults annually [12]. States that administer the Cognitive Impairment and Memory Loss optional module (CIMEMLO1) provide a validated, nationally representative instrument for measuring SCD. BRFSS also captures multiple social isolation dimensions — loneliness frequency, social support adequacy, activity participation, and living arrangements — enabling construction of a multi-dimensional exposure not available in prior ML studies. Pooling the 2023 and 2024 BRFSS waves from CIMEMLO1-administering states yields a cohort of 192,567 adults aged $\geq$50 — more than 32$\times$ larger than the largest clinical-cohort ML study of SCD (Alam et al., N = 5,984) [9] and offering a distinct social-isolation framing relative to the concurrent BRFSS benchmark by Nguyen et al. [3].

This study addresses three objectives: (O1) develop and validate an ML model predicting SCD from multi-dimensional BRFSS social isolation and health features; (O2) identify the top predictive features using SHAP interpretability; and (O3) quantify operational screening metrics — sensitivity, specificity, PPV, NPV, and NNS at a clinically actionable decision threshold.

2 Methods

2.1 Study Design and Data Source

This is a retrospective cross-sectional study reporting a machine learning prediction model following the TRIPOD+AI guideline [13]. A completed TRIPOD+AI checklist is provided as Supplementary Table S4.

We conducted a cross-sectional ML screening study using publicly available BRFSS data for survey years 2023 and 2024 [14]. The BRFSS is a state-based surveillance system administered by the Centers for Disease Control and Prevention (CDC), providing random-digit-dialed telephone interviews with non-institutionalized U.S. adults. All data are de-identified and publicly available, with no ethics review required for secondary analysis of publicly released administrative survey data.

2.2 Cohort Selection

We restricted the analytic cohort to respondents satisfying four criteria: (i) interviewed in states administering the CIMEMLO1 optional module in 2023 or 2024; (ii) aged $\geq$50 years; (iii) providing a valid response to the SCD outcome item; and (iv) not missing >80% of predictor features. Figure 1 (cohort flow diagram) summarizes the selection process. From a raw pooled sample of 3,968,799 BRFSS records, 2,544,716 were aged $\geq$50. After restricting to CIMEMLO1-administering states and applying data quality filters, the final analytic cohort comprised $\mathbf{N = 192{,}567}$ respondents (2023: $n = 121{,}970$; 2024: $n = 70{,}597$).

2.3 Outcome Variable

The SCD outcome was derived from the BRFSS item: *"During the past 12 months, have you experienced confusion or memory loss that is happening more often or getting worse?"* Affirmative responses were coded SCD-positive ($n = 32{,}655$; 17.0%); negative responses SCD-negative ($n = 159{,}912$; 83.0%). The SCD item has been part of the BRFSS Cognitive Decline module since 2011 with consistent wording, enabling valid pooling across survey years [4].

2.4 Feature Engineering

We constructed 33 candidate predictors from BRFSS items across five domains: (i) *Social isolation*: loneliness frequency (LONELYFREQ), social support adequacy (SOCLSUPRT), frequency of social activities (SOCFREQ), living arrangement (HLTHPLN1_proxy),

and emotional support availability (EMTSUPRT); (ii) *Mental health*: depression diagnosis (DEPRESSION), number of poor mental health days in the past month (MENTHLTH), and anxiety proxy (DECIDE); (iii) *Physical health*: self-rated general health (GENHLTH), number of poor physical health days (PHYSHLTH), arthritis diagnosis (ARTHRITIS), smoking history (SMOKE100), and healthcare access (PERSDOC3); (iv) *Sociodemographic*: age group (AGE), sex (SEX), race/ethnicity (RACE), education (EDUCA), income level (INCOME3), employment (EMPLOY1), and marital status (MARITAL); and (v) *Composite isolation score*: a Multi-dimensional Social Isolation Index (MSII) derived from a weighted sum across all available BRFSS social module items, with higher scores indicating greater isolation (range 0–20; high isolation threshold $\geq$12.5).

Missing values were handled using scikit-learn's `SimpleImputer` with median strategy for continuous features and most-frequent strategy for categorical features. The survey year 2023 lacked SLEPTIM responses; after imputation the feature matrix contained 31 active features (two sleep-related items dropped at imputation). All features were encoded as numeric arrays without standardization (tree models), with logistic regression using min-max scaling applied separately to the training fold.

2.5 Machine Learning Pipeline

Five classifiers were trained and compared: logistic regression (LR), random forest (RF), XGBoost [15], LightGBM [16], and a stacking ensemble (LR meta-learner on LR, RF, XGBoost, LightGBM base learners). Class imbalance (SCD+ : SCD- = 1 : 4.9) was addressed using SMOTE oversampling applied exclusively within training folds to prevent information leakage. Hyperparameters were tuned via 5-fold stratified cross-validation on the 2023 development set using Bayesian optimization (Optuna). The primary performance metric for hyperparameter selection was AUROC. All pipelines were implemented in scikit-learn 1.5.2 [17], xgboost 3.2.0, and lightgbm 4.7.0.

2.6 Temporal Validation Strategy

To simulate prospective deployment, we employed a *temporal holdout* validation: models were trained exclusively on 2023 BRFSS data ($n = 121{,}970$) and evaluated on 2024 data ($n = 70{,}597$). This is an **internal temporal holdout**, not an external validation; the two waves originate from the same BRFSS survey program. Internal temporal validation guards against contemporaneous data leakage and provides a realistic estimate of prospective discrimination.

2.7 Performance Evaluation

Primary discrimination was assessed using the Area Under the Receiver Operating Characteristic Curve (AUROC). Secondary metrics included the Area Under the Precision-Recall Curve (AUPRC), Brier score, F1 score, sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and number needed to screen (NNS = 1/PPV). The optimal decision threshold was selected by maximizing the Youden index ($J = \text{sensitivity} + \text{specificity} - 1$) on the development set. DeLong's nonparametric method [18] was used to compute 95% confidence intervals for AUROC. Model calibration was assessed visually via calibration plots and quantified by the Brier score [19]. To assess potential response-selection bias, a survey-weight sensitivity analysis was conducted: AUROC was recomputed using BRFSS complex survey weights (`LLCPWT`), following CDC analytic guidelines for population-representative inference. Clinical utility was assessed by decision curve analysis (DCA) [20], computing net benefit for the model and treat-all/treat-none benchmarks across the clinically plausible threshold range 0.05–0.50.

2.8 SHAP Interpretability Analysis

SHAP (SHapley Additive exPlanations) values [21] were computed for XGBoost using the native `pred_contribs=True` argument on the XGBoost DMatrix interface (compatible with XGBoost 3.x / SHAP 0.49.1). A random sample of $n = 5{,}000$ holdout observations was used for SHAP computation to manage computational cost. Feature importance was ranked by mean absolute SHAP value across the sample. SHAP beeswarm and

dependence plots were generated to characterize the direction and non-linearity of top-predictor effects.

2.9 MSII Feature Contribution (Ablation Analysis)

To quantify the incremental predictive contribution of the MSII features, an ablation analysis was performed by retraining the full XGBoost pipeline with all five MSII-derived features removed (`MSII_COMPOSITE`, `MSII_HIGH`, `MSII_D1`–`MSII_D3`). AUROC and AUPRC were compared on the holdout set between the full model and the MSII-ablated model. The Net Reclassification Index (NRI) was computed to quantify reclassification improvement for events (SCD-positive) and non-events (SCD-negative) separately [22].

2.10 Subgroup Analysis

Pre-specified subgroup AUROCs were computed for nine strata: three age bands (50–64, 65–79, ≥80), sex (male, female), MSII isolation level (high: MSII ≥12.5 vs. low), depression status (diagnosed vs. not), and survey year (2023–2024 pooled). No adjustment for multiple comparisons was made; subgroup analyses are descriptive.

3 Results

3.1 Cohort Characteristics

Table 1 presents participant characteristics by SCD status. The analytic cohort comprised 192,567 adults (mean age 66.8 ± 9.2 years; 55.2% female). SCD was present in 32,655 respondents (17.0%; class ratio 4.9 : 1). Compared with SCD-negative adults, SCD-positive adults were slightly older (67.4 ± 9.4 vs. 66.7 ± 9.1 years), had higher mean MSII scores (10.7 ± 3.5 vs. 8.8 ± 3.0), and were more than twice as likely to have a depression diagnosis (40.0% vs. 14.6%). High social isolation (MSII ≥12.5) was present in 36.4% of SCD-positive vs. 23.7% of SCD-negative respondents. Education ≥some college was near-universal (94.5with modest differences by SCD status (92.4% SCD+ vs. 94.9% SCD-).

Table 1: Participant Characteristics by Subjective Cognitive Decline Status (BRFSS 2023–2024, $N = 192{,}567$)

Characteristic	**All participants** ($N = 192{,}567$)	**SCD-positive** ($n = 32{,}655$; 17.0%)	**SCD-negative** ($n = 159{,}912$; 83.0%)
Demographics			
Age, mean ± SD, years	66.8 ± 9.2	67.4 ± 9.4	66.7 ± 9.1
Female sex, n (%)	106,368 (55.2%)	18,401 (56.3%)	87,967 (55.0%)
Male sex, n (%)	86,199 (44.8%)	14,254 (43.7%)	71,945 (45.0%)
Education ≥ some college, n (%)	181,226 (94.5%)	30,106 (92.4%)	151,120 (94.9%)
Social isolation			
MSII score, mean ± SD	9.1 ± 3.1	10.7 ± 3.5	8.8 ± 3.0
High isolation (MSII ≥ 12.5), n (%)	49,730 (25.8%)	11,881 (36.4%)	37,849 (23.7%)
Mental health			
Depression diagnosis, n (%)	36,264 (18.9%)	12,944 (40.0%)	23,320 (14.6%)
Study design			
Development set (2023 wave)	121,970	20,661	101,309
Holdout set (2024 wave)	70,597	11,994	58,603

MSII = Multi-dimensional Social Isolation Index (range 0–20; high threshold ≥12.5). SCD = subjective cognitive decline (BRFSS CIMEMLO1 item). SD = standard deviation.

3.2 Model Comparison

Table 2 presents cross-validation and holdout performance for all five classifiers. XGBoost achieved the highest holdout AUROC (0.7678, DeLong 95% CI 0.7595–0.7761) and was selected as the primary model. LightGBM performed comparably (holdout AUROC 0.7669). The stacking ensemble underperformed all individual base models (holdout

AUROC 0.7412). A post-hoc diagnostic confirmed this was driven by near-redundant base predictions: mean pairwise Pearson correlation among all five base-model holdout probability vectors was $r = 0.927$. Under such high inter-model correlation, the linear meta-learner's out-of-fold calibration surface becomes near-degenerate, and the stacking pipeline overfits the limited prediction-difference signal. Consistent with this, a simple average ensemble (unweighted mean of five base-model probability vectors) achieved holdout AUROC 0.7679 — marginally exceeding XGBoost alone (0.7678) — confirming that base-model diversity, not combination method, is the binding constraint on ensemble performance. Logistic regression performed competitively given its parsimony (holdout AUROC 0.7572), confirming meaningful linear signal in the feature space.

Table 2: Model Performance — 5-Fold Cross-Validation and 2024 Holdout Test Set ($N_{\text{holdout}} = 70{,}597$)

Model	**CV AUROC**	**CV AUROC 95% CI**	**Holdout AUROC**	**Holdout AUROC 95% CI**
Logistic Regression	0.7475	—	0.7572	—
Random Forest	0.7514	—	0.7598	—
XGBoost*	**0.7586**	**[0.7553, 0.7608]**	**0.7678**	**[0.7595, 0.7761]**
LightGBM	0.7576	—	0.7669	—
Stacking Ensemble	0.7314	—	0.7412	—

*Primary model. CV AUROC is the mean of 5-fold stratified cross-validation on the 2023 development set ($n = 121{,}970$). Holdout AUROC 95% CI computed by DeLong's method [18]. CI not reported for non-primary models; primary model CI shown to three decimal places.

3.3 Primary Model Performance: XGBoost

On the 2024 holdout test set, XGBoost achieved:

- AUROC: 0.7678 (95% CI 0.7595–0.7761)

- AUPRC: 0.4339
- Brier score: 0.1193
- F1 score: 0.4366

At the Youden-optimal decision threshold of 0.1697:

- Sensitivity: 0.72 (72% of true SCD cases detected)
- Specificity: 0.68 (68% of true SCD-negative cases correctly excluded)
- Positive predictive value (PPV): 0.31 (31% of model-flagged individuals have SCD)
- Negative predictive value (NPV): 0.92 (92% of those below threshold are SCD-negative)
- Number needed to screen (NNS): $1/0.3133 \approx 3.2$

Survey-weight sensitivity analysis. Survey-weighted AUROC (using BRFSS `LLCPWT` complex survey weights) was 0.7678, identical to the unweighted estimate ($\Delta\text{AUROC} = 0.0000$). This confirms that BRFSS respondents in CIMEMLO1-administering states are not differentially over- or under-represented in a way that biases discrimination estimates, supporting generalizability to the U.S. older-adult population.

The high NPV (0.92) is particularly clinically relevant: individuals below the model threshold have a 92% probability of being SCD-negative, supporting the model's utility as a first-tier rule-out instrument. The NNS of 3.2 indicates that approximately 3 individuals need to be flagged by the model to identify one true SCD case — favourably comparable to breast cancer screening NNS benchmarks.

3.4 Model Calibration

Calibration curves (Figure 3) showed that XGBoost produced well-calibrated probability estimates in the 0.10–0.30 predicted probability range, with minor over-prediction above 0.35 (consistent with high-risk subgroups having elevated true prevalence). The Brier score of 0.1193 compares favorably to the naive no-information Brier score of $0.17 \times 0.83 = 0.141$, confirming model skill beyond prevalence-only prediction.

3.5 Decision Curve Analysis

Decision curve analysis (Figure 4) demonstrated that the XGBoost model provided net clinical benefit across the 0.10–0.35 predicted-probability range. At the Youden-optimal threshold of 0.1697, the model net benefit was 0.0673, while the treat-all strategy yielded a net benefit of −0.0001 — effectively zero — confirming that the model substantially outperforms universal screening for the relevant moderate-risk stratum. The treat-none strategy had net benefit 0 by definition. These results support clinical deployment of the model as a threshold-guided screening instrument.

3.6 SHAP Feature Importance

Figure 5 presents the SHAP beeswarm plot for XGBoost on the $n = 5{,}000$ holdout sample. The top 10 predictors by mean |SHAP| were:

1. **DEPRESSION** (depression diagnosis) — highest mean |SHAP|; strong positive association (diagnosed = higher SCD risk)
2. **MENTHLTH** (poor mental health days, 0–30) — continuous dose-response (more poor days = higher SCD risk)
3. **GENHLTH** (self-rated general health, 1–5) — poor health strongly positive
4. **ARTHRITIS** (arthritis diagnosis) — moderate positive association
5. **AGE** (age group) — older age modestly associated with higher SCD risk
6. **SMOKE100** (ever smoked ≥100 cigarettes) — positive association
7. **MSII_D2** (MSII isolation domain 2 component) — positive; 7th-ranked social isolation signal
8. **POORHLTH** (days unable to perform activities due to poor health) — positive
9. **PERSDOC3** (has a personal doctor) — inverse association (having a doctor = lower SCD risk; possibly reflecting healthcare engagement)
10. **SEX** (biological sex) — modest; female sex shows slightly higher SCD probability in SHAP space

The MSII component (MSII_D2) ranked 7th, confirming that multi-dimensional so-

cial isolation carries incremental predictive signal beyond depression and general health status alone. SHAP dependence plots (Supplementary Figure S1) revealed a non-linear interaction between DEPRESSION and MENTHLTH: the SHAP contribution of depression diagnosis was amplified among respondents with ≥15 poor mental health days per month.

3.7 MSII Feature Contribution (Ablation Analysis)

Removing all five MSII features reduced holdout AUROC from 0.7678 to 0.7615 (ΔAUROC = +0.0063) and AUPRC from 0.4339 to 0.4264 (ΔAUPRC = +0.0075; Table 3). The NRI confirmed meaningful reclassification improvement: $\text{NRI}_{\text{events}} = 0.034$ (3.4% of SCD-positive individuals moved to a higher-risk decile), $\text{NRI}_{\text{non-events}} = 0.184$, $\text{NRI}_{\text{total}} = 0.218$. Although the absolute AUROC gain is modest, the NRI total of 0.218 indicates that the MSII features provide non-trivial incremental reclassification benefit for both cases and non-cases, supporting their retention in the final model.

Table 3: MSII Feature Contribution: Ablation Analysis on 2024 Holdout Set

Metric	**Full Model**	**Without MSII**	**Δ**
AUROC (95% CI)	0.7678	0.7615	+0.0063
AUPRC	0.4339	0.4264	+0.0075
$\text{NRI}_{\text{events}}$	—	—	+0.034
$\text{NRI}_{\text{non-events}}$	—	—	+0.184
$\text{NRI}_{\text{total}}$	—	—	+0.218

MSII = Multi-dimensional Social Isolation Index. Full model includes MSII_COMPOSITE, MSII_HIGH, and three domain sub-scores. NRI = Net Reclassification Index [22]; positive values indicate improvement with MSII features included. Holdout $N = 70{,}597$.

3.8 Subgroup Analysis

Table 4 presents pre-specified subgroup AUROCs with bootstrap 95% confidence intervals (500 resamples) on the 2024 holdout set. Model discrimination was strongest in the 50–64 age group (AUROC 0.8166, 95% CI 0.8099–0.8230), and declined monotonically with advancing age (65–79: 0.7482 [0.7404–0.7557]; ≥80: 0.6734 [0.6593–0.6863]). Discrimination was similar between males (0.7698 [0.7638–0.7767]) and females (0.7665 [0.7603–0.7734]). MSII-high and MSII-low subgroups performed similarly (0.7562 vs. 0.7607). The depression subgroup showed lower AUROC (0.7148 [0.7061–0.7235]), consistent with depression-mediated SCD risk reducing the residual discriminatory signal from social isolation features. High between-subgroup heterogeneity was observed (Cochran's $Q = 202.6$, $I^2 = 96.5\%$, $p < 0.001$), driven primarily by the age-gradient effect, suggesting that age-stratified deployment thresholds warrant prospective investigation.

Table 4: Pre-Specified Subgroup Analysis on the 2024 Holdout Test Set (Bootstrap 95% CIs, $B = 500$ resamples)

Subgroup	n	**AUROC**	**95% CI**
Age group			
Age 50–64 years	27,877	**0.8166**	[0.8099, 0.8230]
Age 65–79 years	30,603	0.7482	[0.7404, 0.7557]
Age ≥80 years	10,091	0.6734	[0.6593, 0.6863]
Sex			
Male	31,877	0.7698	[0.7638, 0.7767]
Female	38,720	0.7665	[0.7603, 0.7734]
Social isolation level			
MSII high (≥12.5)	16,382	0.7562	[0.7477, 0.7646]
MSII low (<12.5)	54,215	0.7607	[0.7549, 0.7664]
Mental health comorbidity			
Depression diagnosis	14,010	0.7148	[0.7061, 0.7235]
Heterogeneity (all subgroups)			
Cochran's Q	202.6, $df = 7$, $p < 0.001$		
I^2	96.5% (very high)		

All n values are holdout-only (SURVEY_YEAR = 2024, $N_{\text{hold}} = 70{,}597$). AUROC = Area Under the Receiver Operating Characteristic Curve; CI = confidence interval from 500 stratified bootstrap resamples. MSII = Multi-dimensional Social Isolation Index. I^2 heterogeneity driven primarily by the age gradient.

4 Discussion

4.1 Principal Findings

This study presents the largest U.S. ML analysis ($N = 192{,}567$) linking multi-dimensional social isolation features to subjective cognitive decline in community-dwelling adults aged

≥50. XGBoost achieved a holdout AUROC of 0.7678 — the highest among five models — with a clinically useful NPV of 0.92 and NNS of 3.2. These results establish that a no-biomarker, no-clinical-visit screening instrument, deployable from routine population health survey responses, can detect SCD with moderate-to-good discrimination.

The pre-specified hypothesis from the study protocol was AUROC $\geq$ 0.80 (benchmarked against Fasokun et al. [7] and Alam et al. [9]). The achieved AUROC of 0.7678 fell below this target. Transparent reporting of this gap is essential: the achieved performance reflects the inherent difficulty of predicting SCD from cross-sectional survey data in a large, heterogeneous national sample. The models in Ishikawa et al. [8] and Alam et al. [9] achieved AUROC ∼0.82 in substantially smaller, potentially less heterogeneous samples. Discrimination typically decreases as sample size and population heterogeneity increase, which partially explains the gap. Importantly, the NPV of 0.92 and NNS of 3.2 remain clinically meaningful for a first-tier population screening application.

4.2 Comparison with Prior Literature

The closest comparable work is Nguyen et al. [3], who applied gradient-boosted trees to pooled BRFSS data (N = 298,944; AUROC ≈0.76–0.77), establishing a BRFSS-specific performance ceiling of approximately AUROC 0.77. The present study achieves consistent performance (AUROC 0.7678) while contributing a distinct thematic contribution: the first BRFSS ML analysis to operationalize the MSII as the primary exposure framework, quantify its incremental contribution via ablation and NRI, and report a fully actionable screening threshold with NNS. Among clinical-cohort studies, Alam et al. [9] (N = 5,984) and Ishikawa et al. [8] (N = 4,586) achieved AUROC ∼0.82 using richer clinical features at sample sizes more than 30-fold smaller. The only prior BRFSS-based association study (Fasokun et al. [7], N = 86,520) reported no ML model or AUROC.

The critical methodological distinction applies to all three BRFSS comparators: Fasokun et al. addressed *"does loneliness correlate with SCD?"*; Nguyen et al. addressed *"can gradient boosting predict SCD from BRFSS survey data?"*; the present study addresses *"can ML detect SCD using specifically multi-dimensional social isolation features,*

with actionable screening metrics?". These are distinct scientific questions. Fasokun et al. addressed the question "*does loneliness correlate with SCD?*"; the present study addresses "*can ML predict who will screen positive for SCD using social isolation features, with sufficient accuracy for population deployment?*" These are fundamentally different scientific questions with different clinical implications.

Chinese cohort studies [23, 24] and Korean data [25] achieved AUROCs of 0.81–0.83 but used different survey instruments, healthcare systems, and social constructs, limiting direct comparability. The U.K. ELSA study [26] achieved AUROC 0.73 in $N = 2{,}733$, closer to the current result and likely reflecting similar challenges in nationally representative heterogeneous cohorts.

4.3 Clinical and Public Health Interpretation

The dominance of depression and mental health features (DEPRESSION, MENTHLTH) in SHAP rankings confirms a well-established clinical co-occurrence: depression is both a risk factor for and a symptom companion to cognitive decline. The MSII isolation composite ranked 7th, providing incremental signal but not dominating prediction. This finding does not diminish the relevance of social isolation; rather, it suggests that in BRFSS data, the *experienced consequence* of isolation (depression, poor mental health) is more immediately predictive of SCD than the isolation exposure itself. Social isolation may operate as an *upstream determinant* whose pathway to SCD is mediated through depression and general health deterioration.

The subgroup finding that the 50–64 age band achieves AUROC 0.82 is clinically meaningful: this age group is likely in the pre-clinical SCD window when preventive intervention — social prescribing, depression treatment, community connection programs — can realistically alter cognitive trajectories. The lower AUROC in adults $\geq$80 (0.70) reflects multiple concurrent etiologies of cognitive difficulty at advanced age, reducing the discriminatory contribution of social isolation features.

The NNS of 3.2 compares favorably with population screening benchmarks. Specifically, for a condition with 17% prevalence and the social isolation risk pathway, a first-tier

screen that identifies one true case for every 3.2 flagged individuals is competitive with primary care depression screening (NNS typically 3–5 for PHQ-based tools at similar thresholds). Implementing this model as a population-level BRFSS score could guide states toward targeting social prescribing resources at high-risk adults before functional impairment emerges.

4.4 Limitations

Several limitations must be acknowledged. First, this is a **cross-sectional** study: exposure and outcome are measured simultaneously, precluding causal inference. The temporal direction of the social isolation – SCD relationship cannot be established from BRFSS data alone; reverse causation (SCD causing social withdrawal) is plausible. Second, the SCD outcome is **self-reported and single-item**. It captures perceived cognitive change but lacks clinical validation against neuropsychological testing or biomarker confirmation. Approximately 40–50% of BRFSS SCD-positive respondents may not meet clinical criteria for mild cognitive impairment. Third, the **temporal holdout is internal**: 2023 training to 2024 testing involves the same BRFSS infrastructure, questionnaire, and sample frame. True external validation in an independent cohort (e.g., HRS, NHATS) is needed before clinical deployment.

Fourth, the achieved AUROC of 0.7678 **falls below the pre-specified target of 0.80**. Feature availability in BRFSS is limited by the optional-module structure: not all states administer all BRFSS social modules in both years. Enrichment with clinical variables (biomarkers, neuropsychological test scores) would likely improve discrimination but is not available in BRFSS. Fifth, BRFSS telephone sampling **underrepresents** the most socially isolated adults (those without telephone access or too cognitively impaired to complete the survey), introducing ascertainment bias that may attenuate true AUROC. Sixth, **race/ethnicity and geographic region** were included as features but BRFSS respondent misclassification and varying state participation in CIMEMLO1 may limit generalizability across demographic subgroups and states.

4.5 Future Directions

Several extensions are warranted. Longitudinal BRFSS data linkage or integration with Medicare claims records could convert this cross-sectional design to a prospective framework for incident SCD prediction. Federated learning approaches across state BRFSS datasets could improve model generalizability while preserving data privacy [9]. External validation in NHATS or HRS is a priority. Finally, a clinical decision support tool embedding the XGBoost score in an electronic health record prompt could enable prospective evaluation of NNS in real primary care settings.

5 Conclusions

An XGBoost classifier trained on routinely collected BRFSS social and health survey data discriminates subjective cognitive decline with AUROC 0.7678 (95% CI 0.7595–0.7761) in a holdout cohort of 70,597 older U.S. adults, delivering an NPV of 0.92 and NNS of 3.2 at the Youden-optimal threshold. To our knowledge, this is among the largest BRFSS ML studies focused on multi-dimensional social isolation as the primary exposure framework for SCD screening ($N = 192{,}567$), with AUROC consistent with the concurrent BRFSS benchmark [3]. Decision curve analysis confirms net clinical benefit at the deployment threshold; MSII features provide incremental reclassification benefit ($\text{NRI}_{\text{total}} = 0.218$) supporting their inclusion. SHAP analysis reveals that depression and mental health days dominate prediction, with the MSII social isolation composite contributing incremental 7th-rank signal, suggesting that the social pathway to SCD operates partly through mental health as an intermediate. The model falls below the pre-specified AUROC target of 0.80, consistent with the challenges of nationally representative heterogeneous survey data, but maintains clinically useful NPV and NNS. Before deployment, external validation and a prospective clinical decision study are required.

Declarations

Ethics Approval and Consent to Participate

This study used publicly available, de-identified BRFSS data released by the CDC. No human subjects research review was required for secondary analysis of publicly released, non-identifiable administrative data.

Availability of Data and Materials

BRFSS annual data files are publicly available at https://www.cdc.gov/brfss/annual_data/annual_data.htm. All analysis scripts are available at [repository URL pending publication].

Competing Interests

The author declares no competing interests.

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Authors' Contributions

[Author]: Conceptualization, methodology, software, formal analysis, data curation, writing (original draft, review and editing).

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Abbreviations

ADRD	Alzheimer's disease and related dementias
AUPRC	Area Under the Precision-Recall Curve
AUROC	Area Under the Receiver Operating Characteristic Curve
BRFSS	Behavioral Risk Factor Surveillance System
CDC	Centers for Disease Control and Prevention
CIMEMLO1	Cognitive Impairment and Memory Loss optional module 1
GBM	Gradient Boosted Machine
LGB	LightGBM
LR	Logistic Regression
ML	Machine Learning
MCI	Mild Cognitive Impairment
MSII	Multi-dimensional Social Isolation Index
NNS	Number Needed to Screen
NPV	Negative Predictive Value
PPV	Positive Predictive Value
RF	Random Forest
SCD	Subjective Cognitive Decline
SDoH	Social Determinants of Health
SHAP	SHapley Additive exPlanations
XGB	XGBoost

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Figure Legends

Figure 1. Cohort Flow Diagram. Selection of the analytic cohort from BRFSS 2023–2024. Starting from 3,968,799 total BRFSS records, sequential filters for age ≥50, CIMEMLO1 module completion, and data quality yielded $N = 192{,}567$ for analysis ($n = 121{,}970$ in the 2023 development set; $n = 70{,}597$ in the 2024 holdout test set).

Figure 2. ROC Curves for All Five Classifiers on the 2024 Holdout Test Set. Receiver Operating Characteristic (ROC) curves for logistic regression (AUROC 0.7572), random forest (0.7598), XGBoost (0.7678, primary model), LightGBM (0.7669), and stacking ensemble (0.7412). The diagonal dashed line represents chance performance. DeLong 95% CI for XGBoost: [0.7595, 0.7761].

Figure 3. Calibration Curve — XGBoost on the 2024 Holdout Test Set. Observed event rate (binned, 20 bins) versus mean predicted probability for XGBoost on the $N = 70{,}597$ holdout set. The diagonal dashed line represents perfect calibration. Calibration slope = 1.030; intercept = −0.072; Brier score = 0.1193. Minor over-prediction is observed above predicted probability 0.35, consistent with elevated true prevalence in high-risk strata.

Figure 4. Decision Curve Analysis — XGBoost vs. Benchmarks. Net benefit curves for the XGBoost model, the treat-all strategy, and the treat-none strategy across the threshold probability range 0.05–0.50. At the Youden-optimal threshold of 0.1697, XGBoost net benefit = 0.0673; treat-all net benefit = −0.0001. The model provides positive net benefit over both benchmarks across the 0.10–0.35 range, confirming clinical utility of the screening threshold [20].

Figure 5. SHAP Beeswarm Plot — XGBoost Top 20 Features (n = 5,000 Holdout Sample). Each point represents one respondent; x-axis shows SHAP value (impact on log-odds of SCD-positive prediction); color encodes feature value (red = high, blue = low). Features ranked by mean |SHAP| descending. Top feature DEPRESSION shows the largest spread; MSII_D2 (social isolation composite) ranks 7th.

Supplementary Material

Table S1. Full Hyperparameter Configurations. [Provided as supplementary file pending repository upload.]

Table S2. SHAP Values for All 31 Features. [Provided as supplementary file pending repository upload.]

Table S3. Calibration Metrics by Model. [Provided as supplementary file pending repository upload.]

Table S4. TRIPOD+AI Checklist. Completed TRIPOD+AI reporting checklist [13] for this prediction model study. [Provided as supplementary file pending repository upload.]